

Euclid's Elements — Book I

Proposition 12

Formal Reconstruction — Technical Documentation

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CGJTEAMLAB

Constructing a perpendicular from a point to a line

1 Introduction

Euclid's Proposition I.12 asks for the construction of a perpendicular to a given straight line from a prescribed point outside that line.

The classical proof uses a circle. A point is chosen on the side of the given line opposite the prescribed point; the circle centered at the prescribed point through that auxiliary point meets the given line in two points. Proposition I.10 bisects the segment joining those intersections, and Proposition I.8 proves that the line from the external point to the midpoint is perpendicular to the given line.

The current Hilbert reconstruction follows a different synthetic route. It constructs a second point on the opposite side of the base so that two distinct points of the base are equidistant from the original point and the constructed point. The base is then treated as a perpendicular-bisector axis. No circle-line intersection theorem is used.

The current production file contains the general construction theorem

```
hilbert_perpendicular_from_point_exists
```

and the public Euclidean theorem

```
euclid_proposition_12
```

which is a thin wrapper around it.

2 The Classical Configuration

Let AB be the given straight line and let C be the prescribed point outside it. Euclid's construction produces a point H on AB such that

$$CH \perp AB.$$

The classical construction is summarized in Figure 1. The initial-data-only state from the preserved Book1R chapter is not shown as a separate panel: no new geometry has yet been created at that stage.

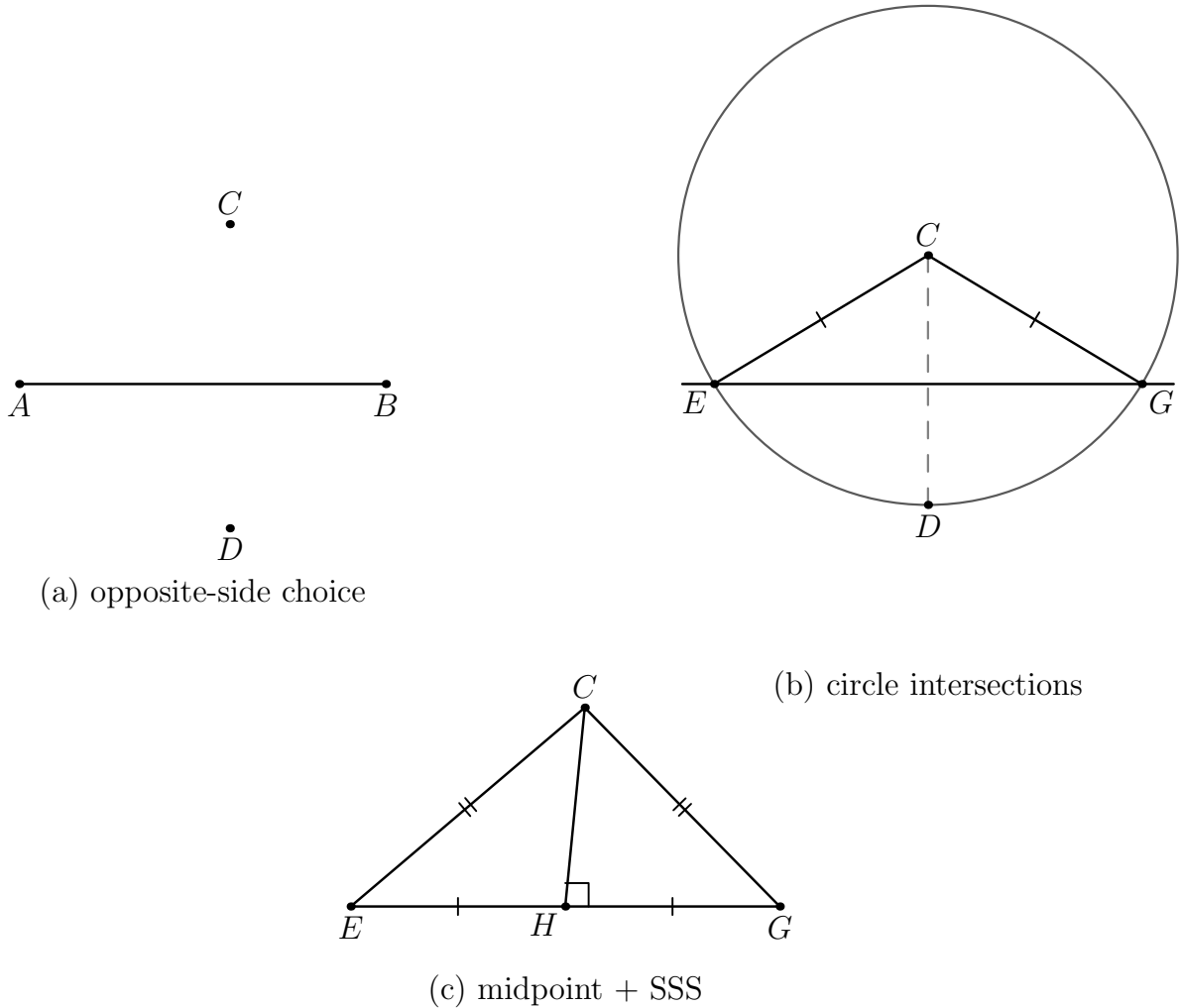


Figure 1: Classical Euclid–Wilkins construction, retaining only genuinely new geometric states. (a) A point D is chosen on the side of the line AB opposite the external point C . (b) The circle centered at C through D meets the given line at E and G , so $CE \cong CG$. (c) Proposition I.10 gives the midpoint H of EG ; after joining CH , $EH \cong HG$, $CE \cong CG$, and the common side CH form the SSS configuration used by Proposition I.8.

2.1 Classical Construction Algorithm

Step 1: given line and external point. The data are the straight line AB and a point C not on it. This is only the initial state and is therefore not assigned a separate figure under the present audit standard.

Step 2: choose an opposite-side point. Choose a point D on the side of AB opposite C ; this is Figure 1(a).

Step 3: draw the circle centered at C . Draw the circle centered at C through D , and let it meet the given line at two distinct points E and G . Then

$$CE \cong CG.$$

This is Figure 1(b).

The existence of the two intersections is a genuine geometric existence issue. The diagram records the configuration after those intersections have been obtained; it does not prove that they exist. This is precisely the kind of hidden classical assumption that should not be replaced by visual intuition.

Step 4: bisect EG . Apply Proposition I.10 to obtain a midpoint H :

$$E - H - G, \quad EH \cong HG.$$

The former midpoint-only snapshot is not retained as a separate panel, because the stable next state immediately joins CH and uses the same points.

Step 5: join CH . The completed proof state is Figure 1(c). The triangles EHC and GHC satisfy

$$EH \cong HG, \quad HC \cong HC, \quad CE \cong CG.$$

By Proposition I.8,

$$\angle EHC \cong \angle GHC.$$

Since E, H, G are collinear, these are congruent adjacent angles and are therefore right angles. Hence

$$CH \perp AB.$$

3 Classical Proof Architecture

The direct classical dependency chain is

$$\begin{array}{c}
\text{choose opposite-side } D \\
\downarrow \\
\text{circle centered at } C \text{ through } D \\
\downarrow \\
E, G \in AB, \quad CE \cong CG \\
\downarrow \\
\text{I.10} \\
\downarrow \\
E - H - G, \quad EH \cong HG \\
\downarrow \\
\text{I.8 / SSS} \\
\downarrow \\
\angle EHC \cong \angle GHC \\
\downarrow \\
CH \perp AB.
\end{array}$$

Thus the proposition-level dependencies of the classical proof include I.10 and I.8. The circle-line intersection itself also carries a classical existence burden not represented by either proposition.

4 Formal Architecture

The current production file imports only

```
CGJteamLab.HilbertBookZero
```

and defines two proposition-specific theorems:

```
hilbert_perpendicular_from_point_exists
euclid_proposition_12
```

The second theorem is a thin wrapper. All geometric work is performed by the first theorem together with reusable Hilbert and Book Zero infrastructure.

The main theorem has the form

```
theorem hilbert_perpendicular_from_point_exists
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B P : Geo.Point)
  (base : Geo.Line)
  (hAB : A != B)
  (hAbase : HilbertIncidence.OnLine A base /\
  (hBbase : HilbertIncidence.OnLine B base)
  (hPbase : not HilbertIncidence.OnLine P base) :
  exists H R : Geo.Point,
    HilbertIncidence.OnLine H base /\
    HilbertIncidence.OnLine R base /\
    HilbertRightAngle Geo R H P
```

Here H is the foot of the required perpendicular. The point R is a second point of the base used to specify the base direction in the predicate `HilbertRightAngle`.

5 Hilbert Reconstruction

The formal construction is summarized in Figure 2.

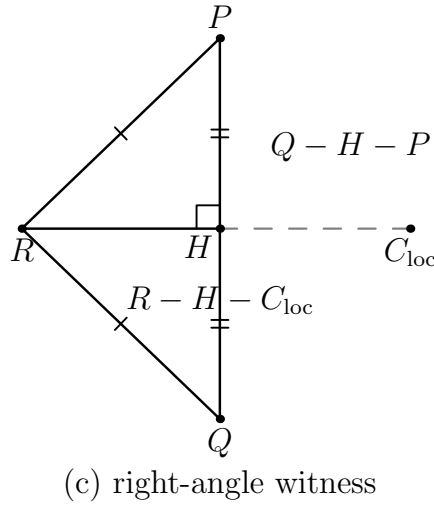
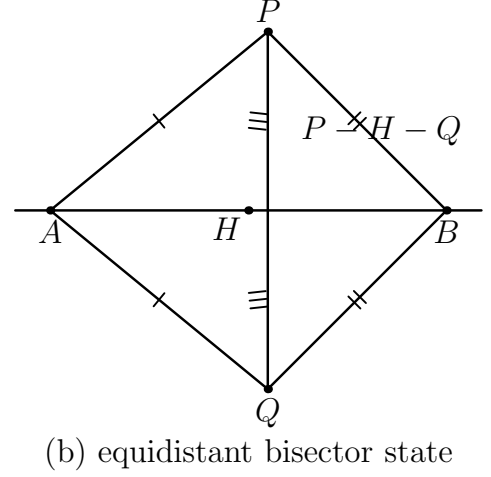
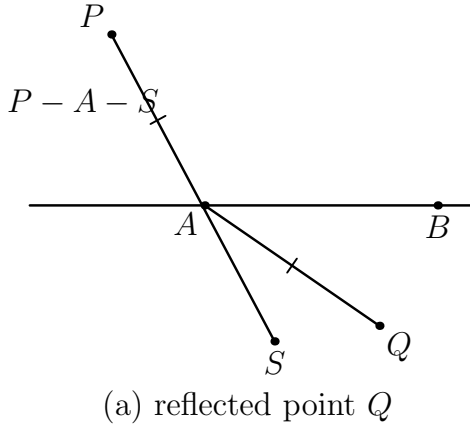


Figure 2: Formal Hilbert construction. (a) The line PA is extended through A to S ; Hilbert angle construction supplies a direction and segment layoff produces Q with $AQ \cong AP$. The intermediate point Q_0 used only to determine that direction is intentionally omitted. (b) SAS gives $BP \cong BQ$; because P and Q are on opposite sides of the base, $P-H-Q$ for an intersection point H , and the reusable perpendicular-bisector theorem gives $PH \cong HQ$. The displayed order of A, H, B is schematic only: the Lean hypotheses assert only that all three lie on the base. (c) After choosing $R \in \text{base}$ with $R \neq H$ and $RP \cong RQ$, the local continuation point C_{loc} with $R-H-C_{\text{loc}}$ packages the final right-angle witness.

5.1 Constructing the opposite-side point Q

First extend PA through A :

$$P - A - S.$$

Since P is off the base and A lies on it, P and S are on opposite sides of the base.

Hilbert angle construction at A produces an auxiliary direction point Q_0 on the side containing S . Segment construction then lays off a point Q on the same ray so that

$$AQ \cong AP.$$

The current proof subsequently transports the side information from Q_0 to Q , obtaining

$$\text{OppositeSide}(P, Q; \text{base}).$$

The point Q_0 is a genuine construction witness, but it is only a direction scaffold. The stable proposition-specific geometry uses Q , and Figure 2(a) therefore does not draw Q_0 as a separate state. The uniqueness field returned by this particular angle-construction call is also not consumed later in the current proof.

5.2 The two equidistance relations

After replacing the auxiliary ray AQ_0 by the ray AQ in the angle congruence, SAS is applied to the triangles APB and AQB . Together with

$$AP \cong AQ, \quad AB \cong AB,$$

it gives

$$BP \cong BQ.$$

The opposite-side relation provides an intersection point H with

$$P - H - Q, \quad H \in \text{base}.$$

The resulting stable state is represented by Figure 2(b).

6 The Perpendicular-Bisector Helper

The reusable theorem

`hilbert_equidistant_line_bisects_segment`

is more structured than the preserved Book1R dependency sketch suggested.

Its inputs include

$$AP \cong AQ, \quad BP \cong BQ, \quad P - H - Q,$$

with A, B, H on the base and P, Q on opposite sides.

First, the current implementation invokes

`hilbert_theorem_17`

and obtains the angle symmetry

$$\angle ABP \cong \angle ABQ.$$

It then performs a genuine positional split on the already existing point H :

The three branches are:

- if $H = A$, the midpoint congruence follows directly from $AP \cong AQ$;
- if $H = B$, it follows directly from $BP \cong BQ$;
- if $H \neq A$ and $H \neq B$, the proof uses `hilbert_collinear_angle_transport`, followed by SAS on the triangles BHP and BHQ .

Thus `hilbert_collinear_angle_transport` is a genuine transitive dependency of I.12, but it is not called directly from `Proposition12.lean`; it occurs only in the generic branch of this reusable Book Zero theorem.

The three cases do not require three separate geometric panels. They do not introduce new points, lines, intersections, or side-of-line configurations; they only determine whether the already constructed intersection H coincides with one of the two base points.

The conclusion is

$$PH \cong HQ.$$

7 The Final Right-Angle Construction

The current main theorem does not stop at the midpoint relation. It defines a local final step for any point R satisfying

$$R \in \text{base}, \quad R \neq H, \quad RP \cong RQ.$$

To package a right angle in the exact formal orientation, Lean extends RH through H to a fresh local point, named C in the source. To avoid collision with the classical external point C in this chapter, Figure 2(c) labels it C_{loc} :

$$R - H - C_{\text{loc}}.$$

SSS is applied to the triangles

$$\triangle RHP \quad \text{and} \quad \triangle RHQ$$

using

$$RH \cong RH, \quad HP \cong HQ, \quad RP \cong RQ.$$

This gives congruent angles at H . Since

$$Q - H - P,$$

vertical-angle congruence converts the result to the adjacent-angle orientation required by

`HilbertRightAngle Geo R H P.`

This local continuation point is a genuine hidden construction and is therefore included in the audited formal diagram.

8 Why the theorem returns a second base point

The main proof finally decides which original base point may be used as R .

It performs

```
by_cases hHA : H = A
```

If $H = A$, then $B \neq H$ and the proof takes $R = B$, using $BP \cong BQ$. Otherwise $A \neq H$ and it takes $R = A$, using $AP \cong AQ$.

This is why the public conclusion existentially returns a second base point R instead of fixing A or B in advance. The two branches do not create distinct final geometric states and therefore do not receive separate panels.

9 Lean Formalization

The public Euclidean theorem is

```
theorem euclid_proposition_12
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B P : Geo.Point)
  (base : Geo.Line)
  (hAB : A != B)
  (hAbase : HilbertIncidence.OnLine A base)
  (hBbase : HilbertIncidence.OnLine B base)
  (hPbase : not HilbertIncidence.OnLine P base) :
  exists H R : Geo.Point,
    HilbertIncidence.OnLine H base /\
    HilbertIncidence.OnLine R base /\
    HilbertRightAngle Geo R H P := by
  exact
    hilbert_perpendicular_from_point_exists
      Geo A B P base hAB hAbase hBbase hPbase
```

Thus `euclid_proposition_12` is a thin interface wrapper. It does not duplicate the construction.

10 Relationship with Earlier Propositions

10.1 Proposition I.8

Euclid uses Proposition I.8 at the final stage of the classical proof. The Hilbert reconstruction also uses an SSS argument, but it is not a call to `euclid_proposition_8`. The SSS configuration is the formal pair of triangles RHP and RHQ .

10.2 Proposition I.10

In Euclid's construction, I.10 bisects EG . The current formal proof does not call I.10. Instead,

$$PH \cong HQ$$

is obtained from `hilbert_equidistant_line_bisects_segment`.

10.3 Proposition I.11

I.11 constructs a perpendicular at a point already on a line. I.12 solves the complementary construction problem, with the prescribed point outside the line. I.11 is not a direct theorem-level dependency of the current I.12 proof.

11 Dependency Summary

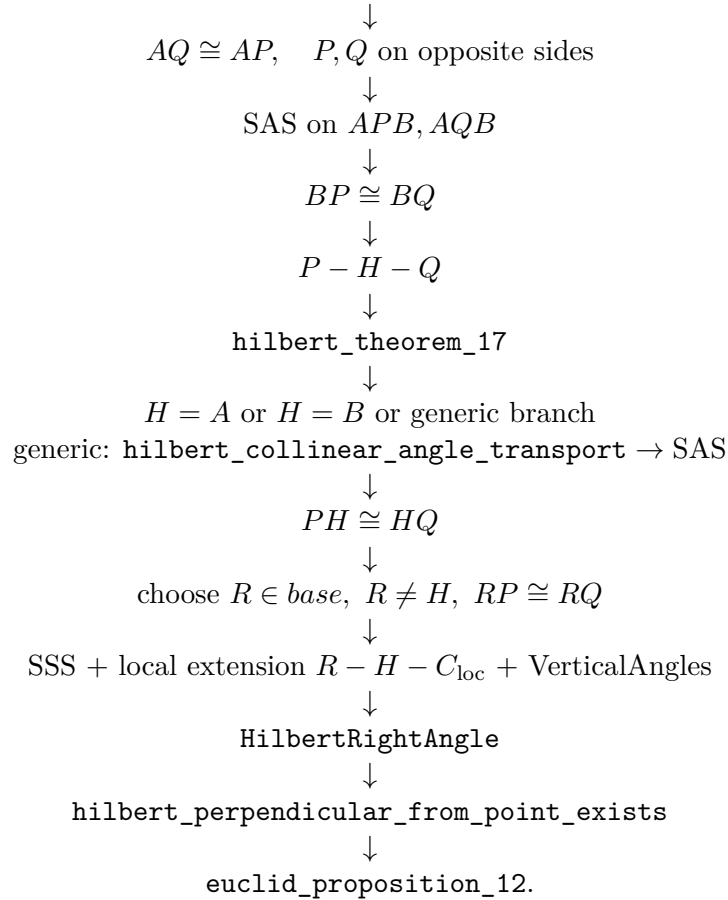
The two architectures should be kept separate.

For Euclid,

$$\text{circle construction} \longrightarrow \text{I.10} \longrightarrow \text{I.8} \longrightarrow \text{I.12}.$$

For the current reconstruction, a more faithful dependency summary is

between extension + Hilbert angle construction + segment layoff



This is the current logical DAG at the level relevant to the proposition and the reusable perpendicular-bisector helper. Internal incidence, ray, side-of-line, and congruence transport lemmas are not expanded into separate diagrammatic states unless they create new geometry.

12 Diagrammatic Audit

The preserved Book1R chapter contained six numbered geometric figures: input data, the opposite-side point, the circle intersections, the completed classical configuration, the formal reflected-point construction, and the formal perpendicular-bisector configuration.

The audit retains all six meaningful geometric stages but reorganizes them into two multi-panel figures.

- The initial-data-only classical figure is removed because it records no new construction.
- The classical opposite-side choice, circle-intersection state, and final midpoint/SSS state become the three panels of Figure 1.
- The two preserved formal states become panels (a) and (b) of Figure 2.
- A new panel (c) records the hidden local continuation $R - H - C_{\text{loc}}$ used by the current source to package the final `HilbertRightAngle` witness.

The formal point Q_0 is intentionally not assigned its own panel: it serves only to choose the construction ray on which Q is subsequently laid off. Likewise, the endpoint-coincidence cases $H = A$ and $H = B$ do not create new geometric objects and are described in the proof DAG rather than duplicated as figures.

A further audit correction concerns Figure 2(b). The current theorem states only that A , B , and H lie on the same base line. It does not establish a betweenness relation among those three points. Their displayed left-to-right order is therefore explicitly schematic and must not be read as a Lean hypothesis.

The final audited set is consequently

2 figures /6 panels .

The organization changes, but the document does not collapse genuinely different geometric states into tactic bookkeeping.

13 Conclusion

Proposition I.12 is an especially clear example of reconstruction rather than literal transcription. Euclid uses a circle, I.10, and I.8. The current Hilbert proof instead constructs an opposite-side point, recognizes an equidistant-line configuration, obtains the midpoint of PQ , and finishes with SSS and vertical angles.

The public Euclidean theorem remains deliberately thin. Its geometric content is carried by `hilbert_perpendicular_from_point_exists` and by reusable Hilbert/Book Zero infrastructure. The audited diagrams now reflect that architecture: the classical circle mechanism and the formal perpendicular-bisector mechanism are visually separated, while every panel records a genuine change of geometry rather than a change of proof state.